

# Increasing Step Rate Reduces Peak and Cumulative Insole Force in Collegiate Runners

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## ABSTRACT

MADDEN, T. S., and D. A. HAWKINS. Increasing Step Rate Reduces Peak and Cumulative Insole Force in Collegiate Runners. *Med. Sci. Sports Exerc.*, Vol. 56, No. 5, pp. 982–989, 2024. Increasing step rate has been proposed to reduce overuse injury risk in runners by reducing the magnitude of musculoskeletal loading. Although estimating structure-specific loads is challenging, during level running, changes in a runner's vertical ground reaction force (vGRF) may provide insight to changes in their musculoskeletal loading. Further, vGRF can be approximated using force sensing insoles. **Purpose:** The primary goal of this study was to examine changes in peak insole force and cumulative weighted peak force (CWPF)/km with increased step rate in collegiate runners. The secondary goal was to determine whether sacral acceleration correlates with insole force when increasing step rate. **Methods:** Twelve collegiate distance runners ran 1000 m outdoors at 3.83 m·s<sup>-1</sup> at preferred and 10% increased step rates while insole force and sacral acceleration were recorded. Cumulative weighted peak force/km was calculated from insole force based on cumulative damage models. The effects of step rate on peak insole force and CWPF·km<sup>-1</sup> were tested using paired *t* tests or Wilcoxon tests. Correlation coefficients between peak axial (approximately vertical) sacral acceleration times body mass and peak insole force were calculated on cohort and individual levels. **Results:** Peak insole force and CWPF·km<sup>-1</sup> decreased ( $P < 0.001$ ) with increased step rate. Peak axial sacral acceleration did not correlate with peak insole force on the cohort level ( $r = 0.35$ ,  $P = 0.109$ ) but did within individuals (mean,  $r = 0.69$ – $0.78$ ;  $P < 0.05$ ). **Conclusions:** Increasing step rate may reduce peak vGRF and CWPF·km<sup>-1</sup> in collegiate runners. Therefore, clinicians should consider step rate interventions to reduce peak and cumulative vGRF in this population. Individual-specific calibrations may be required to assess changes in peak vGRF in response to increasing step rate using wearable accelerometers. **Key Words:** GROUND REACTION FORCE, INJURY, ACCELERATION, WEARABLE SENSORS

Distance running is a popular sport associated with high rates of musculoskeletal injury. In the National Collegiate Athletics Association (NCAA), more than 27,000 distance runners compete on cross-country teams (1). National Collegiate Athletics Association cross-country runners experience 2.85 and 3.44 time loss injuries per 1000 athlete exposures in men and women, respectively (2). Beyond limiting activity and participation, these injuries limit yearly performance (3) and can negatively impact the runner's mental health and quality of life (4).

Increasing step rate has been proposed as an intervention to reduce the risk of running injuries. A higher step rate at a given running speed is associated with reduced bone stress injury risk in collegiate runners (5) and reduced shin injury risk in high school runners (6). Increasing step rate (or equivalently,

decreasing stride length) has been shown to reduce loads at the hip (7,8), knee (7–15), ankle (7,12,16), soleus (10), and Achilles tendon (16) in recreational runners and tibia loads in experienced runners (17).

Whereas estimates of structure-specific loads based on motion capture and musculoskeletal modeling are currently limited to laboratory settings, wearable devices allow estimates of vGRF in natural running environments. Force sensing insoles, which measure the force normal to the plantar surface of the foot, have been used to approximate the peak vGRF during running (18–20). Compared with force instrumented treadmill values, the 200 Hz Loadsol device showed excellent intraclass correlation coefficients (0.95–0.96), mean bias of 0.20 body weight, and standard error of the mean of 0.07 to 0.08 body weight (19). As a practical alternative to laboratory-based assessments, peak insole force may be used to assess the influence of increasing step rate on peak vGRF in settings where estimating structure-specific loads is not yet feasible (e.g., during outdoor running). Further, this approach does not limit vGRF measurements to a treadmill at constrained speed or to a few steps over a force plate in the laboratory, which may not be representative of normal running. Although the vGRF itself does not indicate structural loading, when constrained to level running at submaximal speeds, muscle-tendon and bone forces increase with increasing vGRF (21,22). Therefore, reductions

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TABLE 1. Participant characteristics (mean  $\pm$  SD).

|                   | Female (n = 6)  | Male (n = 6)    |
|-------------------|-----------------|-----------------|
| Age (y)           | 23.7 $\pm$ 2.9  | 23.0 $\pm$ 3.5  |
| Mass (kg)         | 58.8 $\pm$ 6.2  | 74.3 $\pm$ 10.1 |
| Height (cm)       | 165.7 $\pm$ 6.8 | 179.5 $\pm$ 5.7 |
| 5000 m best (min) | 19.4 $\pm$ 1.7  | 15.1 $\pm$ 0.8  |

in vGRF in response to increased step rate during level running may reflect reductions in structure-specific loads.

Overuse injuries are generally understood to occur due to accumulating microstructural damage, which for a single run depends on the number of loading cycles and the load magnitude per cycle. To develop effective training strategies that modify these factors to reduce the risk of overuse injury, it is important to understand the relative weight the number of loading cycles and the load magnitude have on microstructural damage. Fatigue failure behavior of musculoskeletal structures suggests the damage accumulated depends more strongly on load magnitude than the number of loading cycles (23). Therefore, a reduction in load magnitude with increased step rate is likely to reduce the damage accumulated for a given running distance despite the increased number of steps. Metrics of cumulative damage (23) can help test this hypothesis and provide greater insight to potential changes in injury risk than estimating peak vGRF alone.

Inexpensive wearable devices, such as accelerometers, may be able to estimate changes in peak vGRF in response to increasing step rate in the field. The product of body mass and peak axial (approximately vertical) acceleration at the sacrum can provide a simplified estimate of peak vGRF. This method produced a moderate ( $r = 0.64$ ) and significant correlation with force-instrumented treadmill measurements of vGRF in collegiate cross-country runners (24). Therefore, measuring changes in sacral acceleration may provide a less expensive and intrusive approach than laboratory-based assessments or

force sensing insoles to assess changes in peak vGRF when running with increased step rate.

Increasing step rate at constant running speed has been shown to decrease peak vGRF (8,10,25) and insole force (26) in recreational runners, but few studies have evaluated the effect of the intervention on metrics of cumulative damage potentially related to injury risk (17), and no study to date has investigated these effects in collegiate runners specifically. In addition, it is unknown whether sacral acceleration can assess (i.e., correlates with) changes in peak vGRF with increased step rate.

The primary purpose of this study was to examine the changes in peak insole force and cumulative weighted peak force (CWPF) per kilometer, with increased step rate in collegiate distance runners. As a secondary purpose, we also aimed to determine whether peak sacral acceleration correlates with peak insole force when increasing step rate. Our primary hypothesis was that peak insole force and CWPF would decrease with increased step rate. Our secondary hypothesis was that peak axial sacral acceleration would decrease with increased step rate and that peak axial sacral acceleration multiplied by body mass would correlate with peak insole force on cohort and individual levels.

## METHODS

**Participants.** NCAA and club collegiate cross-country runners ( $n = 12$ ) from UC Davis and St. Mary's College of California were recruited (Table 1). Runners competed in events of 1500 m or longer. All participants were uninjured at the time of study, where injury was defined as a reduction in training due to pain. Sample size was determined via an *a priori* power analysis using G\*Power software (Dusseldorf, Germany). The expected effect size was determined from preliminary data of five collegiate or former collegiate distance runners (2 female/3 male). At a constant speed of  $3.83 \text{ m}\cdot\text{s}^{-1}$ , peak insole force decreased by  $0.0908 \pm 0.0849$  body weight

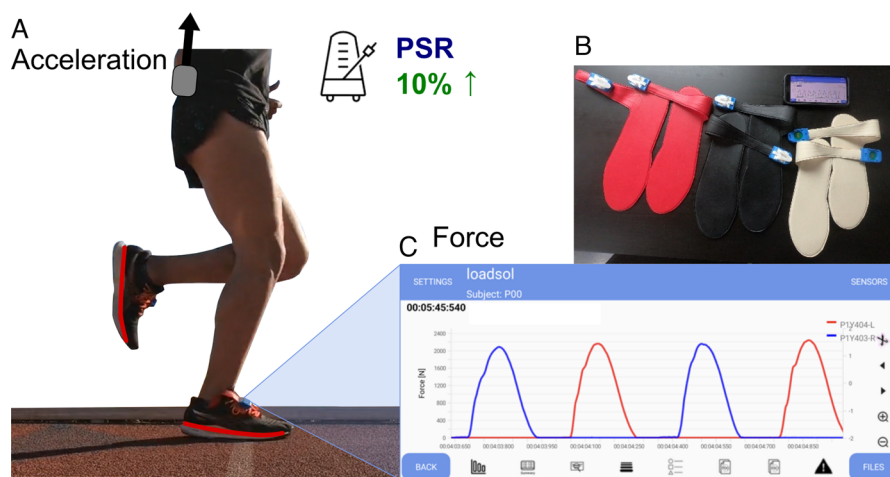


FIGURE 1—Experimental protocol overview. (A) Participants matched steps to an audible metronome set to frequencies of preferred step rate (PSR) and 10% increased step rate. Acceleration was recorded from an IMU placed over the sacrum. (B) Loadsol force sensing insoles recorded insole force data via (C) the Loadsol app on an Android device.

(mean  $\pm$  SD) on average from preferred to 10% increased step rate conditions, resulting in an effect size of 1.07. No sex-based difference in response was observed. With this expected effect size, power of 0.80, and significance level of 0.05, a minimum of nine participants was determined necessary to detect differences in peak insole force with sufficient power. Participants provided written informed consent and the study was approved by the UC Davis Institutional Review Board.

**Experimental protocol.** Participants were fit to wireless force sensing insoles (Novel Loadsol, St. Paul, MN), which measure the normal force under the plantar surface of the foot (Fig. 1B). These sensors were previously validated to measure vGRF accurately and reliably during running (18–20). Three insole sizes were used in this study. Participants were fit to the smallest insole size that covered the entire plantar area of the foot. The insoles were inserted into the shoes over the insoles of the participants' own shoes and calibrated according to manufacturer instructions. The calibration for each sensor was verified with static single leg stance force within 5% of the participant's body weight, measured using an electronic scale. Insole forces were recorded at 200 Hz.

A wearable inertial measurement unit (IMU) (ProMove Mini, Inertia Technology, Enschede, The Netherlands) with an acceleration measurement range of  $\pm 16$  g was mounted on the sacrum (Fig. 1A) using a neoprene belt over the waistband of the shorts and secured as tightly as comfortable. The vertical axis of the IMU coordinate system was aligned with the vertical axis of the trunk. The acceleration along this axis was taken as the vertical sacral acceleration and recorded at 200 Hz.

To provide performance-related context to our results, we also examined the effect of increased step rate on heart rate and perceived exertion. Participants wore a Polar chest strap heart rate monitor (H10, Kempele, Finland) that updated heart rate data at 1 Hz via a Garmin device (Forerunner 230, Olathe, KS). After each trial, participants reported their rating of perceived exertion (RPE) using a modified Borg CR10 scale (27).

To determine each participant's preferred step rate, participants first ran for 400 m around a synthetic outdoor track at their preferred step rate next to a pacer running at  $3.83 \text{ m}\cdot\text{s}^{-1}$ . This speed was chosen to represent the speed used during most training runs for the NCAA distance runners at UC Davis, based on conversations with athletes. Insole force data were transmitted via Bluetooth from the electronics housing, attached

to the laces of the participants' shoes, to an Android device with Loadsol software (Fig. 1C). The preferred step rate was taken as the average step rate calculated using the Loadsol software during the final 30 s of the trial.

Participants then ran 1000 m around the track next to the pacer at  $3.83 \text{ m}\cdot\text{s}^{-1}$  at their preferred step rate and at a 10% greater step rate in random counterbalanced order. One 1000-m trial was completed for each step rate condition. Step rate was enforced using an audible metronome. After acclimating to the specified step rate for 200 m, data were recorded for 1000 m. Trials were accepted if the average speed, calculated from 1000-m time, was within 5% of  $3.83 \text{ m}\cdot\text{s}^{-1}$  and the average speeds of the two trials were within 2.5% of each other. To synchronize the force and acceleration data, participants performed a vertical countermovement jump before and after each trial. Participants rested for a minimum of 3 minutes between trials.

**Data processing.** Force data were imported into MATLAB (R2021b, Natwick, MA) for data processing and analysis using custom MATLAB scripts. Raw force data were left unfiltered (19). Vertical sacral acceleration data were low-pass filtered using a zero-lag 4th order Butterworth filter with a 5-Hz cutoff frequency. This cutoff frequency was found to result in the lowest absolute error between predicted peak vGRF using peak sacral acceleration times body mass and the observed peak vGRF in collegiate cross-country runners (24). Although those results suggested a 10-Hz cutoff frequency yields stronger correlations between peak sacral acceleration times body mass and peak vGRF than a 5-Hz cutoff frequency (24), in the current study, a 5-Hz cutoff frequency resulted in stronger correlations between peak sacral acceleration times body mass and peak insole force. The gravitational component of the vertical sacral acceleration was calculated as the mean static acceleration and subtracted from the acceleration signal during dynamic trials. Force and acceleration data were time synchronized (Fig. 2) via cross-correlation of the peak force and vertical sacral acceleration signals obtained during the countermovement jumps (24). Heart rate data were downloaded from the Garmin device and analyzed in MATLAB.

**Cumulative damage model.** To investigate changes in cumulative loading with increased step rate, we defined a cumulative weighted peak force metric based on fatigue failure behavior of repetitively loaded musculoskeletal structures

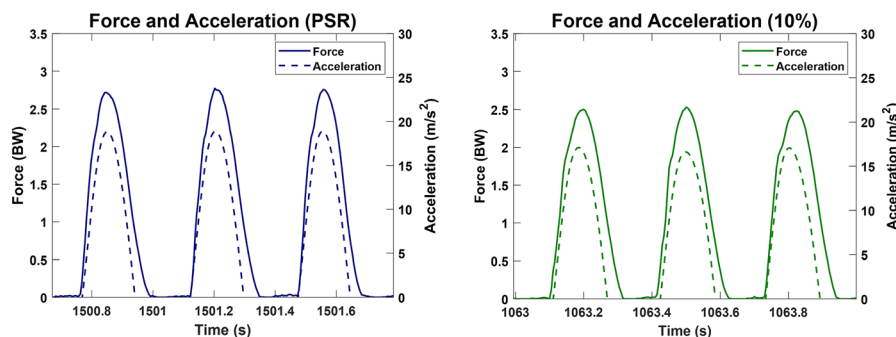


FIGURE 2—Synchronized force and acceleration for preferred step rate (left) and 10% increased step rate (right).

(23). Using the Palmgren-Miner rule (28) and empirical relationships between the number of cycles to failure and peak cyclic stress of musculoskeletal structures loaded to fatigue failure (29–32), cumulative damage (CD) is proportional to the product of the number of steps ( $n$ ) and peak stress per step ( $S_i$ ) weighted by an experimentally derived factor ( $b$ ), summed over  $i$  steps of varying peak stress (23):

$$CD \propto \sum_i n_i S_i^b \quad (1)$$

The constant  $b$  varies by tissue: 4.3 for patellar tendon (29), 6.6 for cortical bone (17,32), and 9.3 for the Achilles tendon (31).

For a specific runner, a metric for cumulative loading to the leg (cumulative weighted peak force, CWPF) during a run can be defined by replacing stress in Eq. 1 with peak vGRF:

$$CWPF = \sum_{i=1}^n F_i^{6.6} \quad (2)$$

where  $n$  is the total number of steps and  $F_i$  is the peak vGRF force of the  $i$ th step for each leg. Given the association between step rate and the risk of bone stress injury in collegiate runners (5), a common (2,5), and debilitating (33) injury in this population, we chose to weight the peak force by a factor of 6.6, although the exact weighting is not critical to the goals of this study. Previous work has shown a relationship between a similar measure of cumulative weighted vGRF, estimated using accelerometry, and injury (34).

**Analysis.** Variables of interest were calculated for each step rate condition. Step rate was calculated using the insole force data. Peak insole force, peak sacral acceleration, impulse, and contact time were analyzed as per-step averages over each 1000-m trial. Peak insole force and peak axial sacral acceleration were determined for each step and these values were averaged over each 1000-m trial. Impulse per step was calculated from the insole force data as the integral of the force-time curve over stance, where stance start and end points were defined as the times at which the slope of the force-time curve exceeded  $2500 \text{ N}\cdot\text{s}^{-1}$  and fell below  $-2500 \text{ N}\cdot\text{s}^{-1}$ , respectively. This method was based on that of Seiberl et al. (20), who used a  $\pm 1500 \text{ N}\cdot\text{s}^{-1}$  threshold to identify stance start and end points with insole force data recorded at  $100 \text{ Hz}$  in participants running at  $2.2\text{--}3.3 \text{ m}\cdot\text{s}^{-1}$ . For the faster running speed in this study, we used a threshold of  $\pm 2500 \text{ N}\cdot\text{s}^{-1}$  to account for elevated forces near a clear stance phase. Contact time in seconds was calculated from the stance start and end times for each step. Percent of time in ground contact was calculated as the sum of the left and right ratios of contact time to stride time, which was determined using the stance start points of consecutive ipsilateral steps. Cumulative weighted peak force per kilometer was calculated for each 1000-m trial according to Eq. 2 using the peak insole force. Values for each parameter were averaged across right and left legs for each trial. Finally, the heart rate for each step rate condition was averaged over the final 60 seconds of each trial.

For data visualization, insole force was calculated at each relative point of the stride cycle. Data were normalized to

101 points for each stride cycle (stance start to subsequent stance start). At each point in the stride cycle, the mean insole force of each participant was calculated across right and left sides for each trial. The means across participants were then calculated for each step rate condition.

The effect of step rate on each variable of interest was tested for significance using matched pairs tests. Data were evaluated for normality using Q-Q plots and Shapiro–Wilk tests with a significance level of 0.05. For normally distributed data, differences in variables between step rate conditions were tested for significance using paired t-tests with a significance level of 0.05. For variables that violated the assumption of normality (CWPF per kilometer and RPE), differences were tested for significance using Wilcoxon signed rank tests with significance level of 0.05. Acceleration data were analyzed for 11 participants due to lost signal for one participant. Heart rate data were analyzed for 11 participants due to a lost connection between devices for one participant.

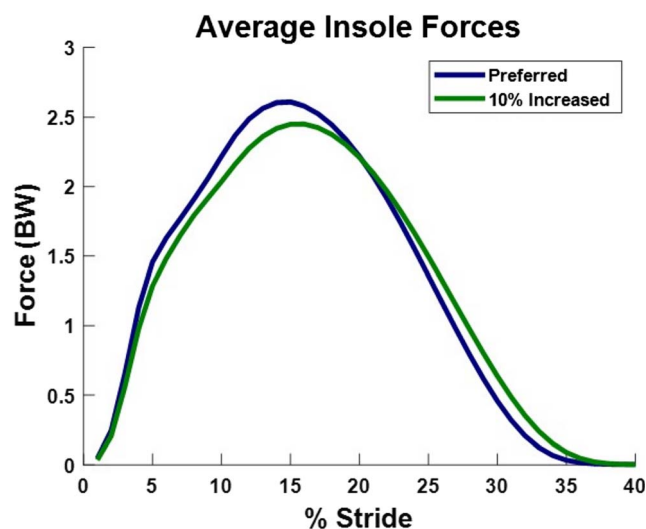
Correlations between peak insole force and peak sacral acceleration multiplied by body mass were evaluated on the cohort level and within participants. The peak accelerations corresponding to the peak forces per step were extracted by identifying the peak accelerations nearest in time to the peak forces. Peak acceleration per step was then multiplied by participant body mass. The Pearson correlation coefficient,  $r$ , was calculated for the cohort between average peak sacral acceleration multiplied by body mass and average peak insole force. These values were normalized by body weight before calculating the correlation coefficient. We also evaluated within-participant correlations between peak sacral acceleration multiplied by body mass and peak insole force. Left and right sides were evaluated separately. The average correlation coefficients were then calculated across participants. A strong correlation was defined as  $r \geq 0.80$ , moderate as  $0.50 \leq r < 0.80$ , and weak as  $r < 0.50$ .

## RESULTS

Peak insole force was lower in the increased step rate condition compared with the preferred step rate condition (Fig. 3). Average peak insole force, CWPF, and peak sacral acceleration significantly decreased with increased step rate (Table 2). Average peak insole force decreased by  $0.16 \pm 0.07 \text{ BW}$  (mean  $\pm$  SD), which represents a  $5.9 \pm 2.7\%$  decrease in peak force with increased step rate relative to preferred step rate. The decrease in average peak insole force translated to an average  $26.1 \pm 12.6\%$  decrease in CWPF.

The increased step rate resulted in significant differences in all other parameters tested (Table 2). Participants increased step rate by an average of  $9.7 \pm 0.9\%$  (mean  $\pm$  SD) relative to the preferred step rate. Running speed showed a slight ( $<1\%$ ) but significant increase with increased step rate. Impulse and contact time per step also decreased with increased step rate; however, percent of time in ground contact increased. RPE and heart rate both significantly increased with increased step rate. Consistent with the decrease in peak force,





**FIGURE 3**—Average insole forces over the stride cycle for preferred and 10% increased step rate conditions. Values were averaged across all participants.

peak vertical sacral acceleration decreased by  $5.8 \pm 4.6\%$  with increased step rate.

Peak sacral acceleration multiplied by body mass did not correlate with peak insole force on the cohort level but did within individuals. On the cohort level, the correlation between peak sacral acceleration multiplied by body mass and peak insole force was weak and not statistically significant ( $r = 0.35$ ,  $P = 0.109$ ; Fig. 4). Within individuals, relationships varied by participant and sides within participants (Fig. 5). Although all statistically significant ( $P < 0.05$ ), individual and side-specific correlations ranged from weak ( $r = 0.19$ ) to strong ( $r = 0.92$ ). Left and right correlations were on average moderate (Table 3).

## DISCUSSION

Our primary objective was to examine changes in peak insole force and CWPf per kilometer with increased step rate in collegiate runners. Our primary hypothesis was that average peak insole force and CWPf per kilometer would decrease with increased step rate. In support of our hypothesis, average peak insole force and CWPf per kilometer significantly decreased while running at a 10% increased step rate relative to preferred, indicating that increasing step rate results in decreased peak vGRF and cumulative weighted peak vGRF per kilometer. These results are consistent with peak insole force changes (26) and peak vGRF changes (8,10,25) reported for recreational runners. A decrease in CWPf per kilometer may suggest a decrease in cumulative damage in bone structures for which loads are reduced, such as the tibia (17). Since other structures exhibit similar fatigue failure behavior (29–31), it may also indicate a decrease in cumulative damage of other musculoskeletal structures for which structure-specific loads are reduced. Therefore, the observed decreases in peak insole force and CWPf over 1000 m of outdoor running provide ecologically valid evidence that a 10% increase in step rate may

reduce the peak and cumulative weighted vGRF, which may also suggest decreases in bone and other structure-specific cumulative damage and relative decreases in structure-specific injury risk. Although we tested peak insole force and CWPf at an arbitrary 10% increase in step rate, our results suggest that a more subtle increase in step rate may still elicit meaningful reductions in peak and cumulative weighted vGRF.

As a secondary objective, this study also evaluated the use of sacral acceleration to assess changes in peak insole force (i.e., explain variation in peak force). Our secondary hypothesis was that peak axial sacral acceleration would decrease with increased step rate and that peak axial sacral acceleration multiplied by body mass would correlate with peak insole force on cohort and individual levels. Consistent with decreased peak force, peak axial sacral acceleration significantly decreased with increased step rate. However, peak axial sacral acceleration multiplied by body mass did not significantly correlate with peak insole force on the cohort level ( $P = 0.109$ ). In contrast, on the individual level, correlations between peak axial sacral acceleration multiplied by body mass and peak insole force were all significant and on average moderate ( $r = 0.69$ – $0.78$ ). In partial support of our hypothesis, these results suggest that sacral acceleration can be used to assess changes in peak insole force not on a group level but only on an individual level. Day et al. (24) showed peak vGRF correlates with sacral acceleration multiplied by body mass across speeds in collegiate runners. Differences between this correlation and the results of the present study may be because of the smaller range in peak insole force observed when increasing step rate (0.8 BW) compared with the range in peak vGRF observed across speeds (1.2 BW, 24) and errors between peak vGRF and peak insole force (18–20). To assess changes in peak vGRF (and thus CWPf) with increased step rate, this approach may require individual-specific calibrations in which an initial session with force sensing insoles, force plates, or an instrumented treadmill is used to generate a model to predict the individual's peak insole force or vGRF given peak sacral acceleration. Practitioners can then use that relationship to estimate peak vGRF in response to increased step rate or other interventions in the field. A secondary analysis of our results revealed that changes in estimated peak insole force with increased step rate correlated strongly with changes in measured values only when individual-specific calibrations

**TABLE 2.** Variables for each step rate condition and changes from preferred to 10% increased step rate (mean  $\pm$  SD).

| Variables                                          | Step Rate Condition |                   | Change             | P      |
|----------------------------------------------------|---------------------|-------------------|--------------------|--------|
|                                                    | Preferred           | 10% Increased     |                    |        |
| Speed ( $\text{m}\cdot\text{s}^{-1}$ )             | $3.79 \pm 0.07$     | $3.83 \pm 0.06$   | $0.04 \pm 0.04$    | 0.010  |
| Step rate ( $\text{min}^{-1}$ )                    | $176 \pm 8$         | $193 \pm 8$       | $17 \pm 1$         | <0.001 |
| Peak insole force (BW)                             | $2.64 \pm 0.23$     | $2.49 \pm 0.24$   | $-0.16 \pm 0.07$   | <0.001 |
| CWPf ( $10^5 \text{ BW}^{6.6}$ )                   | $2.85 \pm 1.64$     | $2.14 \pm 1.38$   | $-0.71 \pm 0.55$   | <0.001 |
| Impulse ( $\text{BW}\cdot\text{s}$ )               | $0.352 \pm 0.026$   | $0.318 \pm 0.023$ | $-0.034 \pm 0.008$ | <0.001 |
| Contact time (s)                                   | $0.228 \pm 0.014$   | $0.217 \pm 0.014$ | $-0.011 \pm 0.004$ | <0.001 |
| % time in ground contact                           | $66.8 \pm 2.9$      | $69.7 \pm 3.1$    | $2.9 \pm 0.9$      | <0.001 |
| Peak acceleration ( $\text{m}\cdot\text{s}^{-2}$ ) | $15.2 \pm 1.8$      | $14.3 \pm 1.8$    | $-0.9 \pm 0.7$     | 0.002  |
| RPE                                                | $3.3 \pm 1.4$       | $4.6 \pm 1.6$     | $1.3 \pm 1.5$      | 0.016  |
| Heart rate (bpm)                                   | $161 \pm 14$        | $164 \pm 12$      | $3 \pm 3$          | 0.015  |

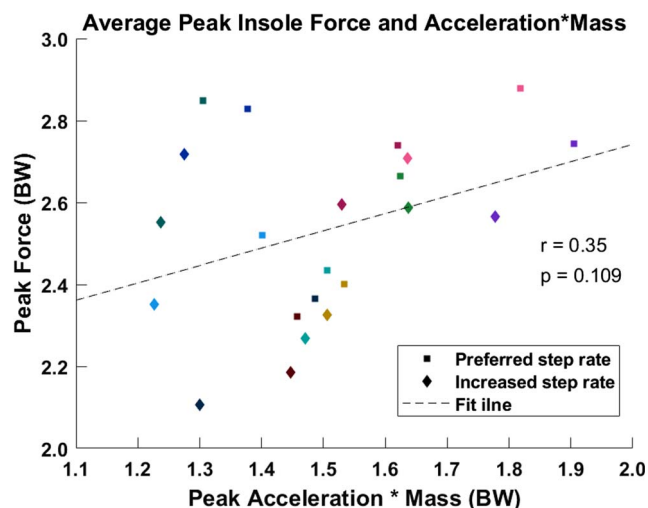


FIGURE 4—Average peak insole force and peak acceleration multiplied by body mass, each normalized by body weight. Each color represents one participant ( $n = 11$ ).

were applied ( $r = 0.92$ ,  $P < 0.001$  with vs.  $r = 0.52$ ,  $P = 0.101$  without). Although individual-specific calibrations may allow the use of sacral acceleration to assess changes in peak vGRF, the requirement of an initial insole or laboratory-based calibration session limits the large-scale feasibility of this approach. Furthermore, if the goal is to estimate vGRF magnitude (e.g., to make comparisons between individuals or against normative data), more accurate prediction models with additional variables (35,36) are likely needed.

Increasing step rate also resulted in changes in impulse and spatiotemporal measures. Impulse per step significantly decreased with increased step rate, likely due in part to decreased peak force and contact time. A decrease in vertical impulse suggests a lower vertical take-off velocity and thus a lower center of mass vertical excursion (8). These changes contribute to the decrease in step time that is required to increase step frequency. Although contact time per step significantly decreased, the percent of stride time in ground contact significantly increased with increased step rate. On average, percent of time in ground contact was 2.9% longer when increasing step rate, which is consistent with Heiderscheit et al. (8). Interestingly, a shorter percent of time in ground contact was consistent with greater peak vGRF and associated with a higher risk of bone stress injury in collegiate runners (37). However, the difference in percent of time in ground contact observed here is smaller than the differences between clusters of different injury incidences (37).

RPE and heart rate significantly increased with increased step rate. These changes could be a result of increased mental focus and/or increased metabolic energy consumption to run at a higher step rate. The increase in heart rate contrasts with previous studies showing no significant increase in heart rate and metabolic cost with 10% increase in step rate (38) and no significant increase in metabolic cost with 8% increased step rate (39). These differences may be due to differences in competition level of the samples. Hamill et al. (38) studied healthy

male adults who showed a U-shaped trend for heart rate and oxygen consumption where the minimum occurred at a step rate greater than preferred. The collegiate runners in this study may have preferred step rates that are already at or near optimum for metabolic cost such that increasing step rate may increase the metabolic energy consumption at a given speed. Given the increases in RPE and heart rate with increased step rate in collegiate runners observed here, increasing step rate should be prescribed only for “easy” runs in which the performance for that run is not the primary goal.

Slight changes in running speed and individual differences in responses may have contributed to the large variability in CWPf changes with increased step rate ( $SD = 12.6\%$ ). Although speed was controlled to within 2.5% between trials, there was a small ( $<1\%$ ) and significant increase in speed with increased step rate. Preferred step rate tends to increase with speed (21). Therefore, participants may have tended to run slightly faster in attempt to maintain a more comfortable step rate for their running speed. Despite a slightly faster speed on average with increased step rate, all participants showed reduced peak insole force and CWPf and results are consistent with studies that constrained speed using instrumented treadmills (8,10,25).

The peak and cumulative weighted force measurements of this study were limited to insole force used to approximate the vGRF. Therefore, we cannot infer relative changes in the risk of injury for specific musculoskeletal structures. However, in level running, changes in vGRF reflect changes in structure-specific forces (21,22). Other studies have shown decreased hip extensor (10), patellar tendon (10), patellofemoral (10,11,13,14), tibiofemoral (9,14,15), tibia (17), soleus (10), and Achilles tendon loads (16) with increased step rate. Therefore, the decrease in CWPf observed here may indicate a decrease in cumulative damage for those structures. However, increasing step rate may increase hip flexor muscle forces during early swing and hamstring muscle forces during late swing (10) and therefore may not be appropriate for reducing the risk of all injuries. For some of the most common running injuries including Achilles

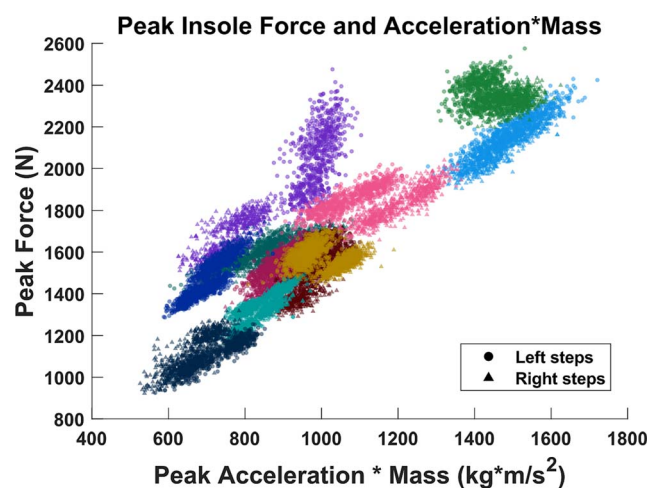


FIGURE 5—Peak insole force and peak acceleration multiplied by body mass for left and right side steps. Each color represents one participant ( $n = 11$ ).

TABLE 3. Participant-specific correlations between peak sacral acceleration multiplied by body mass and peak insole force.

| Participants  | Correlation Coefficient |                 |
|---------------|-------------------------|-----------------|
|               | Left                    | Right           |
| 1             | 0.57                    | 0.76            |
| 2             | 0.89                    | 0.89            |
| 3             | 0.92                    | 0.89            |
| 4             | 0.87                    | 0.89            |
| 5             | 0.32                    | 0.60            |
| 6             | 0.19                    | 0.40            |
| 7             | 0.87                    | 0.89            |
| 8             | 0.87                    | 0.90            |
| 9             | 0.62                    | 0.75            |
| 10            | 0.55                    | 0.77            |
| 11            | 0.91                    | 0.85            |
| Mean $\pm$ SD | 0.69 $\pm$ 0.26         | 0.78 $\pm$ 0.16 |

tendinopathy, patellar tendinopathy, and tibial stress fracture (40), increasing step rate may be an appropriate intervention. Clinicians should consider interventions to increase step rate in runners susceptible to such injuries (e.g., those with a frequent injury history or those returning from injury) and implement interventions gradually to allow adaptation to potential shifts in load to different musculoskeletal structures.

The CWPf metric in this study (Eq. 2) represents a surrogate of cumulative damage in bone (e.g., the tibia). When peak force was weighted by factors reflecting the fatigue failure behavior of Achilles tendon (9.3; 31) or patellar tendon (4.3; 29), CWPf per kilometer decreased by 36.2% ( $P < 0.001$ ) and 16.8% ( $P < 0.001$ ) on average, respectively, which is consistent with the 26.1% decrease specific to bone. This metric only represents a surrogate of cumulative damage for a single loading bout and thus does not consider damage repair and adaptation of musculoskeletal structures. More detailed, and structure-specific models of damage repair and adaptation would need to be incorporated to infer differences in cumulative damage over multiple training sessions.

The results of this study are specific to level track running at 3.8 m·s<sup>-1</sup>. It is unknown whether the same effects of increasing step rate on normal or vGRF are observed during graded running. In addition, peak vGRF does not adequately reflect structure-specific loads during graded running (22). Although

our results are limited to a track surface, similar results were observed for running on treadmill (8,10,25) and road (26) surfaces in recreational runners. The speed used in this study represents easy run speeds and the majority of runs in the population of NCAA runners. It is unknown whether increasing step rate at faster speeds would have the same effects on biomechanical and physiological parameters.

The results of this study are also specific to the acute effects of increasing step rate in collegiate runners; the longitudinal effects of increasing step rate in this population are unknown. Future work should investigate the longitudinal effects of a training protocol to increase step rate on vGRF, structure-specific loads, and performance-related metrics in collegiate runners. Prospective studies should be conducted to determine whether interventions targeting an increase in step rate reduce the risk of running injuries.

## CONCLUSIONS

In conclusion, our results provide ecologically valid evidence that increasing step rate results in decreased peak vGRF and CWPf per kilometer in collegiate distance runners. The decrease in CWPf, which represents a potential surrogate of the general damage accumulated in musculoskeletal structures for a given training session, may reflect relative decreases in structure-specific injury risk. Therefore, clinicians should consider interventions to increase step rate in runners susceptible to injury (e.g., those with a history of frequent injuries or those returning from injury). Increases in RPE and heart rate suggest a greater mental focus and/or metabolic cost with increased step rate in this population, therefore interventions to increase step rate should be constrained to easy running. Correlation results suggest individual-specific calibrations are required to use sacral acceleration as a surrogate to assess changes in peak vGRF in response to increased step rate or other training or footwear interventions in the field.

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